LESSON 7.10
MULTIPLYING INTEGERS USING INTEGER CHIPS

LESSON INTRODUCTION

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

LEARNING TARGETS

- I can use integer chips to multiply integers.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- How do you use integer chips to model multiplication of integers?
- How do you determine the sign of the product when multiplying integers?

LESSON NARRATIVE

In this lesson, students use integer chips to model integer multiplication. Students work with integers that have absolute values less than or equal to 12. The lesson progresses from representing multiplication using integer chips to multiplying integers with opposite signs before closing with multiplying two negative integers. The focus is on students observing a pattern in the signs of the products while making sense of algebraic expressions using integer chips.

COMPONENT SUMMARY

7.10.1 WARM-UP *(~5 MIN.)*

Solve a puzzle by adding and subtracting integers

7.10.2 EXPLORATION *(10 MIN.)*

Use integer chips to model multiplication of integers

7.10.3 COLLABORATION *(5 MIN.)*

Explore using zero pairs to multiply integers with opposite signs

7.10.4 EXPLORATION *(5 MIN.)*

Use integer chips to multiply integers with opposite signs

7.10.5 TRY IT *(5 MIN.)*

Generalize understanding of multiplying integers with opposite signs

7.10.6 EXPLORATION *(5 MIN.)*

Explore how zero pairs are used to multiply integers that are both negative

7.10.7 TRY IT *(5 MIN.)*

Practice multiplying integers that are both negatives

7.10.8 WRAP-UP *(~5 MIN.)*

Determine sign of products of integers

RESOURCES

GLOSSARY

Key Terms:

- Commutative property of multiplication

Additional Terms:

- N/A

MATERIALS

- Integer chips
- (Optional) Colored pencils

ADDITIONAL PREPARATION

- If you do not have integer chips, you can print and cut them out from a free online template, have students draw pictorial representations, or use virtual manipulatives.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle multiplying integers if they are not fluent in multiplying whole numbers.
- Students may have trouble conceptually understanding why the sign is negative if the integers have opposite signs and positive if both integers are negative.
- Students may not recall the commutative property of multiplication.

THINGS TO CONSIDER

- While there are several components, each is meant to be brief so that students have an opportunity to explore, practice, and formalize what they are learning throughout the lesson in chunks.
- Monitor for students who are correctly modeling the problems with the integer chips and have them share their work with the class.
- If you do not have integer chips, students can draw pictorial representations or use virtual integer chips.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

7.10.3 Collaboration provides opportunities for students to initially try the problems independently and then discuss their work with a partner. The discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 7.10.4 and 7.10.6 Exploration activities, students have opportunities to represent multiplying integers using multiple representations. Students follow a concrete-pictorial-abstract progression as they work with written descriptions of real-world scenarios, integer chips, and numeric expressions.

DIFFERENTIATE INSTRUCTION

Consider using our On-Ramp to 6th Grade tool to identify and provide remediation to students who would benefit from extra practice multiplying whole numbers. If students have delays or deficiencies in their fine motor skills, they can work with a peer tutor or use virtual manipulatives instead of concrete integer chips in the 7.10.5 and 7.10.7 Try It activities. If students finish 7.10.4 or 7.10.6 Exploration activities early, ask them to create example multiplication problems with their partners, swap with other pairs, and solve each other's problems.

7.10.1 WARM-UP: ADDING AND SUBTRACTING

Component Overview: *Students solve an addition and subtraction of integers puzzle requiring them to fill in six blanks with the digits one to six to make specific descriptions of the values true.*

ENRICHMENT

- Is there a way to rewrite the last two expressions?
- What do you notice about the first and third expressions?

7.10.1 Warm-Up: Adding and Subtracting

1. **Place a digit in each blue box so that the expressions in the first row have the same value and the expression in the second row has the largest value. Use only digits 1 to 6, at most one time each.**
   Answers will vary.

Description of the Values	Expressions	Digits to Choose From
Same value	$-\boxed{6} + \boxed{3}$ $-\boxed{2} - \boxed{1}$	1 2 3 4 5 6
Largest value	$-\boxed{4} - -\boxed{5}$	

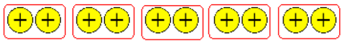
Other possible solutions:

$-6 + 1$
$-2 - 3$
$-4 - (-5)$

$-6 + 2$
$-1 - 3$
$-4 - -5$

7.10.2 Exploration: Using Integer Chips to Represent Multiplication

1. The addition expression $7+7+7$ can be written as $3 \cdot 7$. This is because repeated addition of integers can be expressed as
   - o subtraction.
   - o multiplication.
   - o division.
2. The expression 5×2 can be interpreted as adding 5 groups of 2 or adding 2 groups of 5.
3. Use integer chips to represent one of these interpretations.
   Recall that ⊖ $= -1$ and ⊕ $= +1$.
   Draw a picture of your representation.
   Adding 5 groups of 2



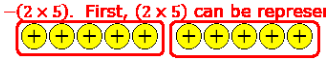
   OR Adding 2 groups of 5

4. Describe how the integer chips represent the value of the expression 5×2.
   Answers will vary. In the representation, 5×2 is the sum of the 5 groups of 2 positive integer chips. The sum is 10. Likewise, 2×5 is the sum of the 2 groups of 5 positive integer chips. The sum is also 10.
5. The expression $5 \times (-2)$ can be interpreted as adding 5 groups of -2 or adding -2 groups of 5.
6. Explain which interpretation of $5 \times (-2)$ you think would be easier to represent using integer chips.
   Answers will vary. "Adding 5 groups of -2" is easier to represent using integer chips because this expression means adding 5 groups of 2 negative integer chips. The other interpretation, "adding -2 groups of 5", may be interpreted as the opposite of 2 groups of 5 or $-(2 \times 5)$, which is more difficult to represent using integer chips.
7. Discuss your choice with your partner. Then, work together to use integer chips to represent one of the interpretations. Draw a picture of the representation.
   Answers will vary. 5 groups of -2

   OR
   -2 groups of 5

8. Describe how the integer chips represent the value of the expression $5 \times (-2)$.
   $5 \times (-2)$ means adding 5 groups of negative 2.

   The representation shows 10 negative integer chips. Therefore, the integer chips represent their sum.
   OR The other interpretation, "adding -2 groups of 5", may be interpreted as the opposite of 2 groups of 5 or $-(2 \times 5)$. First, (2×5) can be represented as



   The opposite of this would then be

9. What is the value of the expression $5 \times (-2)$?
   -10

Commutative Property of Multiplication
$a \times b = b \times a$

The **commutative property of multiplication** states that the order in which multiplication is performed does not change the value of the expression.

10. Using the commutative property of multiplication, what is the value of the expression $^{-}2 \times 5$?
    -10

7.10.2 EXPLORATION: USING INTEGER CHIPS TO REPRESENT MULTIPLICATION

Component Overview: *Students relate prior knowledge of how to represent multiplication as repeated addition to modeling integer multiplication with integer chips. Students work with integers with the same and with different signs. The commutative property of multiplication is also revisited.*

ENRICHMENT

What do you notice about adding 5 groups of 2 and adding 2 groups of 5? Why might this be important?

TEACHER MOVES

Encourage mathematical discourse throughout the Exploration, as students have an opportunity to both verbally process and independently or collaboratively write explanations or descriptions in questions 4, 6, 7, and 8.

Monitor for student responses that demonstrate understanding or possible misconceptions while students are exploring. Elicit these responses when facilitating a short debrief after the activity.

ENRICHMENT

What do you notice about the value of both expressions?

7.10.3 COLLABORATION: ARE ZERO PAIRS NECESSARY FOR MULTIPLICATION?

Component Overview: *In this Collaboration, students will discuss and question how zero pairs can be used in modeling multiplication of integers with opposite signs.*

TEACHER MOVES

Think-Pair-Share

Consider structuring this activity with a Think-Pair-Share routine. Students will be collaborating throughout the Collaboration. Providing an opportunity to independently (think) synthesize their findings from 7.10.2 Exploration while analyzing the work of others will produce stronger responses while they refine their answers with a partner (pair) before answering questions 1 and 2 and sharing whole group (share).

7.10.3 Collaboration: Are Zero Pairs Necessary for Multiplication?

Rashaad and Sebastián were discussing how to represent $^{-}2 \times 5$ using integer chips.

Sebastián said, "I don't know how you can have negative 2 groups."

Rashaad said, "I agree. We were able to represent $5 \times {}^{-}2$. Why can't we figure out how to represent the equivalent expression $^{-}2 \times 5$?"

Sebastián then asked, "When we added and subtracted using integer chips, we used zero pairs. Do you think using zero pairs would help?"

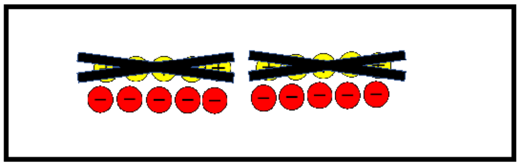
1. Discuss with your partner Sebastián's question, "Do you think using zero pairs would help?"
2. Briefly explain your answer to his question. Draw integer chips as part of your explanation if that is helpful.
Answers will vary. If 2×5 means add 2 groups of 5 positive integer chips, then -2×5 could mean subtracting 2 groups of 5 positive integer chips. Since there is nothing to subtract the 2 groups of 5 from, we can represent the "nothing" using 2 groups of 5 zero pairs. Then, we can subtract the 2 groups of 5 positive integer chips, and the result is 2 groups of 5 negative integer chips, or -10.

7.10.4 Exploration: Using Integer Chips to Represent Multiplication with Negative Groups

In the subtraction problem $5 - 2$, the -2 implies that 2 should be taken away from 5. In the expression $^{-}2 \times 5$, the $^{-}2$ can be interpreted as removing 2 groups of 5.

Use the steps below to discover how to represent $^{-}2 \times 5$ using integer chips.

Step 1: First, use your integer chips to create 2 groups of 5 zero pairs. Draw the 2 groups of 5 zero pairs in the box below.



Step 2: Recall that the expression $^{-}2 \times 5$ is interpreted as removing 2 groups of 5. From your integer chips, remove 2 groups of 5 positive integer chips. In the box above, show removing 2 groups of positive 5 integer chips by crossing them out.

The value of the expression $5 \times (-2)$ is -10. Due to the commutative property of multiplication, $^{-}2 \times 5$ is expected to have the same value.

Does the number of chips that remain match the expected value of $^{-}2 \times 5$?

○ Yes,
○ No, **the value of $^{-}2 \times 5$ is -10 according to the model.**

7.10.4 EXPLORATION: USING INTEGER CHIPS TO REPRESENT MULTIPLICATION WITH NEGATIVE GROUPS

Component Overview: *Students use integer chips to multiply integers with opposite signs. Students use zero pairs to model the expressions and interpret negative groups as removing groups. Check for understanding with a whole-class discussion before proceeding to the next lesson component.*

QUESTIONING

If students finish early, ask them to explain why the commutative property holds for integer multiplication. Scaffold students who are unsure with prompts such as “What other operation does the commutative property hold for?”

TEACHER MOVES

Monitor for student responses that reflect the discourse from 7.10.2 Exploration. “The visual model represents the opposite of two groups of five.”

7.10.5 TRY IT: MULTIPLYING INTEGERS WITH OPPOSITE SIGNS

Component Overview: *Students practice using integer chips to multiply integers with opposite signs. Question 2 demonstrates how when multiplying integers with opposite signs, the result is always negative. The commutative property justifies why the order of the signed integers does not matter.*

TEACHER MOVES

Prompt students to articulate the verbal description of each expression. “How would you describe this mathematical expression using the word ‘opposite’ and ‘group’?”

Expression 1: “Adding six groups of negative 2.”

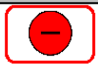
Expression 2: “Adding four groups of negative 1.”

7.10.5 Try It: Multiplying Integers with Opposite Signs

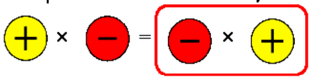
1. Complete the table.

Expression	Visual Model Using Integer Chips	Product
$6 \times (-2)$		-12
$(4)(^{-}1)$		-4

2. Use the pattern of the multiplication in the table to complete the rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
	

3. Multiplication is commutative, so ...



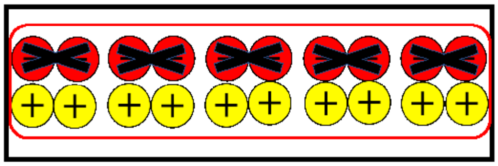
7.10.6 Exploration: Multiplying a Negative Integer by a Negative Integer

1. Complete the statement.

   The expression, $(-5)(-2)$, indicates that 5 groups of -2 should be removed.

2. Use the steps below to discover how to represent $(-5)(-2)$ using integer chips.

Step 1: First, use your integer chips to create the set of zero pairs that will be needed to find $(-5)(-2)$. Draw the groups of zero pairs in the box below.



Step 2: Remove 5 groups of 2 negative integer chips.

Step 3: Remove any remaining zero pairs, if needed.

3. Complete the statement.

   The value of expression $(-5)(-2)$ is 10.

4. With your partner, discuss whether or not you think the product when multiplying two negative values will always be positive. Explain your thinking.
   Answers will vary. I think the product will always be positive because when we create the zero pairs, we are removing groups of negative integers, which only leaves us with positive integers left.

7.10.6 EXPLORATION: MULTIPLYING A NEGATIVE INTEGER BY A NEGATIVE INTEGER

Component Overview: *Students explore how zero pairs are used to multiply integers that are both negative. Students learn to think of a negative integer multiplied by a negative integer as removing negative integers from the zero pairs that represent the expression. They generalize about whether the sign will always be positive.*

ENRICHMENT

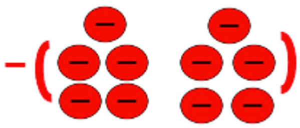
Based on the previous activities, what do you predict the sign will be when multiplying a negative integer by another negative integer?

QUESTIONING

Challenge students to find more than one way to represent multiplying two negative integers with integer chips accompanied by a different verbal expression, building on the use of the word "opposite" from previous activities.

Option 1: As seen and described in the answer key.

Option 2: "The opposite of two groups of negative five."



DIFFERENTIATE INSTRUCTION

Consider polling the class on question 4. Then ask a few students to justify their responses. After the discussion, repoll the class to see if any students changed their minds.

7.10.7 TRY IT: MULTIPLYING A NEGATIVE INTEGER BY A NEGATIVE INTEGER

Component Overview: *Students continue practicing multiplying two negative integers using integer chips and determine that the sign of the product will always be positive. Give students the option to use concrete integer chips, draw pictorial representations of integer chips, or determine the product without integer chips.*

QUESTIONING

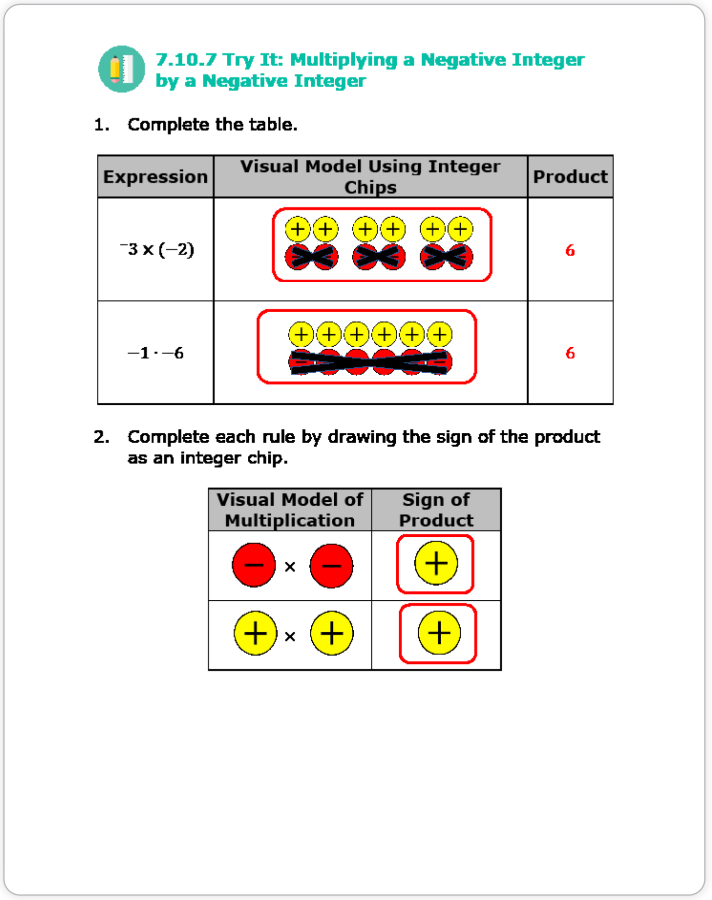
Ask students who get done early to examine some problems where three integers are multiplied.

- (2)(-3)(4)= -24
- (-2)(-3)(4)=24
- (-2)(-3)(-4)=-24
- (2)(3)(4)=24
- (-2)(3)(4)=24

Prompt students to form a generalization or rule for determining the sign of a product of any number of integers. The goal is for students to see that if there is an even number of negative factors, the product is positive. However, if there is an odd number of negative factors, the result will be negative.

ENRICHMENT

When you are multiplying integers, how do you know when you will get a positive or a negative product?



7.10.7 Try It: Multiplying a Negative Integer by a Negative Integer

1. Complete the table.

Expression	Visual Model Using Integer Chips	Product
$^{-}3 \times (-2)$		6
$-1 \cdot -6$		6

2. Complete each rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
− × −	+
+ × +	+

7.10.8 Wrap-Up: Cool Down

1. Complete each rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
+ × −	−
− × +	−
− × −	+
+ × +	+

7.10.8 WRAP-UP: COOL DOWN

Component Overview: *Students determine the signs of products of integers. Consider using data from this formative assessment to determine which students need remediation before the next lesson.*

DIFFERENTIATE INSTRUCTION

Prompt students to provide means of proving their answer. Students can choose to reference an example from the lesson and writing the number next to the assigned row, or students could produce their own examples using integer chips or write a numeric expression.